Conversational AI for Sustainable Agriculture

GreenFieldData PhD Position L

Purpose of the assignment

This assignment is intended to provide a homogeneous reference for all candidates while leaving enough space for creativity, technical judgement, and personal initiative. Candidates may use any tools they normally use, including AI-based assistants, provided that they can explain, justify, and critically discuss the content they submit. The objective is not to verify whether the text was produced without assistance, but to understand how much control the candidate has over the proposal, the technical choices, the risks, and the scientific reasoning.

Context

The PhD position in Conversational AI for Sustainable Agriculture aims to design natural-language interfaces over robotic and analytical farming systems. These interfaces should enable non-expert users to command, monitor, and adapt agri-robots, IoT infrastructures, and streaming analytics workflows. The position includes challenges such as heterogeneous IoT data streams, precision pesticide spraying, human-in-the-loop supervision, semantic grounding, feedback loops, latency-aware edge-cloud deployment, and responsible use of LLM-based conversational interfaces.

Assignment

Prepare an original work plan of **approximately 4–6 pages** explaining how you would shape the PhD work, starting from the published title and abstract of the position. The work plan should not be a generic statement of interest. It should show how you understand the research problem, how you would transform it into concrete tasks, and how you would evaluate progress during the PhD.

Your proposal may focus on one concrete scenario, for example soil-moisture monitoring, precision pesticide spraying, robot failure diagnosis, conversational orchestration of field missions, or another scenario that you justify as relevant to the position.

Suggested structure

1. Research framing: explain the problem you would address, why it matters for sustainable agriculture, and how conversational AI can support non-expert users without replacing human judgement.
2. Proposed PhD tasks: describe 3–5 research or engineering tasks you would design and implement. For each task, explain its objective, input data, expected output, and relation to the overall PhD objective.
3. Workflow and infrastructure: explain how you would organise data, models, experiments, and system components. You may discuss Metaflow or another workflow/MLOps approach to support reproducibility, artifact versioning, auditability, and deployment across local, edge, and cloud environments.
4. Methods and experiments: describe the methods you would explore, such as data engineering, NLP, LLM-based interaction, semantic grounding, robot/IoT integration, machine learning, human-in-the-loop evaluation, or uncertainty estimation. Explain what experiments you would conduct and what datasets or simulated environments you would use.

5. Evaluation criteria and metrics: define how you would evaluate the work technically and from the user perspective. Consider accuracy, latency, robustness, data quality, usability, explainability, safety, trust calibration, and environmental or operational constraints.
6. Responsible AI safeguards: explain how you would prevent unsafe automation, hallucinated recommendations, over-trust in AI outputs, poor explainability, biased assumptions, or inappropriate delegation of decisions to an LLM. Include safeguards such as human confirmation, uncertainty reporting, traceability, role-based permissions, fallback procedures, and explicit limits on what the system is allowed to do.
7. Reflection on tool use: briefly state whether and how you used AI-based tools to prepare the assignment. Explain how you checked, corrected, and took responsibility for the final content.

Expected outcome

The expected outcome is a concise, original, and technically grounded work plan. The answer should show that the candidate can reason about the PhD as a research project, not only as a coding task. It should also demonstrate that the candidate can combine technical competence with critical thinking, responsible AI awareness, and the capacity to explain choices clearly.

References

Use at least three bibliographical references and declare the citation style used, such as APA, IEEE, or MLA. At least one reference should address workflow orchestration, Metaflow, MLOps, or reproducibility; one should address smart agriculture, IoT, agricultural robotics, or crop sensing; and one should address conversational AI, responsible AI, human-centred AI, or human-in-the-loop systems. Briefly justify why each reference is relevant to your proposal.

Evaluation dimensions

Dimension	What will be discussed during the interview
Research understanding	Capacity to identify a concrete research problem and connect it to the PhD objectives.
Technical judgement	Capacity to propose appropriate methods, workflows, data, infrastructure, and experiments.
Creativity and initiative	Capacity to propose an original, feasible, and intellectually engaging direction.
Responsible AI awareness	Capacity to identify risks, safeguards, limits, and human-centred design principles.
Ownership of the proposal	Capacity to explain, defend, revise, and critically discuss the submitted work, including any AI-assisted parts.